Computer Networks

Subject: Computer Science | NoteShare Library

COMPUTER NETWORKS

1. OSI MODEL
7. Application
6. Presentation
5. Session
4. Transport
3. Network
2. Data Link
1. Physical

2. TCP/IP
4. Application (5-7)
3. Transport
2. Internet
1. Network Access

3. TCP vs UDP
TCP: connection, reliable, ordered
UDP: connectionless, fast

TCP handshake: SYN, SYN-ACK, ACK

4. IP ADDRESSING
IPv4: 32-bit
IPv6: 128-bit
Classes A-E
Private: 10.x, 172.16-31.x, 192.168.x
CIDR: 192.168.1.0/24
NAT, DHCP, DNS

5. PROTOCOLS
HTTP/HTTPS (80/443)
FTP (21), SFTP (22)
SMTP (25), POP3 (110), IMAP (143)
DNS (53), DHCP (67/68)
SSH (22), Telnet (23)

6. ROUTING
Static vs Dynamic
RIP, OSPF, BGP
Dijkstra, Bellman-Ford

7. DEVICES
Hub, Switch, Router, Bridge, Gateway
Firewall, Modem, Access Point

8. WIRELESS
802.11 (WiFi)
WEP, WPA, WPA2, WPA3
Bluetooth, Cellular